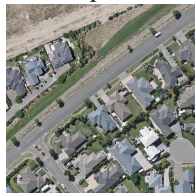
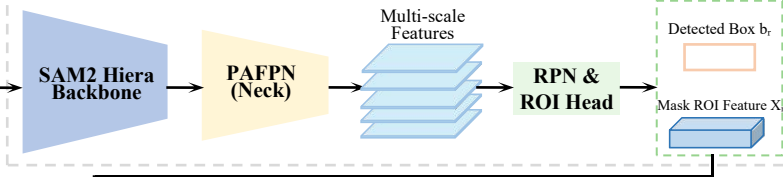


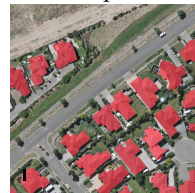
Input



Candidate Generation and Image Encoding



Output



Coarse Shape Prompt Mining(CSPM)

Mask ROI Feature X_r Detected Box b_r

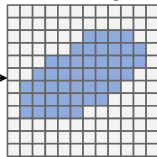
Coarse Mask Decoder

Conv Block $\times 2$ Upsample $\times 2$

Conv Block

Upsample $\times 2$

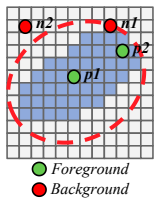
Conv

Coarse Mask Logits C_r 

Resize

Paste

Adaptive 2P2N Sampler

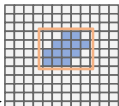


Sparse Prompts

Detected Box b_r

2P2N Points

Dense Prompts



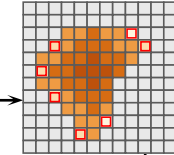
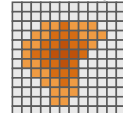
Uncertainty-Guided Decoder-Tail Refinement (UDR)

Image Embeddings

SAM2 Prompt Encoder

SAM2 Mask Decoder

High Low

Mask Logits Y_r^0 Mask Token t_r Local Feature z Position x_i, y_i Uncertainty $|Y_{rs}^0|$ Native Logit Y_{rs}^0 Refined Mask Logits Y_r^1  $\{\delta_i\}$ Φ_i

MLP

 $2 \times \text{Tanh}$

FC

GELU

FC

LayerNorm

Mask

Post-processing